

UNIT-3 | Lecture 2

SQL Queries, Constraints, Set Operations & Nested Queries

Coverage: Hours 3–4

Learning Objectives

Understand SQL query structure and common constraints.

Use UNION, INTERSECT, EXCEPT, and nested queries.

Form of SQL Query

```
SELECT column_list  
FROM table_list  
WHERE condition;
```

Common additional clauses: **GROUP BY**, **HAVING**, and **ORDER BY**.

SQL Constraints

PRIMARY KEY: uniquely identifies a row and cannot be NULL.

FOREIGN KEY: references a key in another table.

NOT NULL: disallows NULL values.

UNIQUE: prevents duplicate values.

CHECK: enforces a condition.

DEFAULT: provides a default value.

UNION, INTERSECT & EXCEPT

UNION: combines compatible query results and removes duplicates by default.

INTERSECT: returns rows common to both query results.

EXCEPT: returns rows from the first result that are absent from the second; exact support varies by DBMS.

Set-operation queries require compatible column counts and compatible data types.

Nested Queries

A **subquery** is a query inside another SQL statement.

```
SELECT Name FROM Student  
WHERE Student_ID IN  
(SELECT Student_ID FROM Enrollment);
```

Subqueries are commonly used with **IN**, **EXISTS**, **ANY**, **ALL** and comparison operators.

Key Takeaway

SQL provides declarative queries and constraints for practical relational database management.